Tutorial 07

ENVX2001 – Applied Statistical Methods

Semester 1

Welcome to week 7!

This week's lecture introduced you to Simple- and Multiple- Linear Regression. For both of these regression techniques we defined the key steps in model fitting and how to interpret model output.

In this tutorial, we will explore key differences between simple linear regression (SLR) and multiple linear regression (MLR), and we will learn how to fit these models in R. We will also discuss how to test assumptions and evaluate model fit for both types of regression.

You will learn how to:

1. Distinguish key similarities and differences between simple linear regression (SLR) and multiple linear regression (MLR) models
2. Implement the fitting process for each type of model
3. Test assumptions and evaluate model fit of SLR and MLR models.

Simple linear regression vs Multiple Linear Regression

Table 1: Similarities and differences between MLR and SLR models

Element	Simple Linear Regression (SLR)	Multiple Linear Regression (MLR)
Number of predictors	1	2 or more
Model form	$Y = \beta_0 + \beta_1 x + \epsilon$	$Y = \beta_0 + \beta_1 x_1 + +\beta_2 x_2 + ... + \beta_n x_n + \epsilon$
How to fit in R	`lm(y ~ x, data = dataset)`	`lm(y ~ x1 + x2 + ... + xn, data = dataset)`
Assumptions	L.I.N.E.	L.I.N.E., collinearity
Model evaluation	r^2, residual plots	Adjusted r^2, residual plots, VIF
Conclusion	"as x increases, y also increases/decreases by β_1 units"	"as x_1 increases, y also increases/decreases by β_1 units, holding all other predictors constant"

Setting up for our analysis

Packages

This lab uses `tidyverse`, `readxl`, `moments`, `psych`, and `performance`. Install any you are missing by running the following **in the console**:

```
CODE
install.packages(c("tidyverse", "readxl", "moments", "psych", "performance"))
```

If a package is already installed, R will simply reinstall it — no harm done.

The data

For both examples we will use the *trees* dataset, which is an in-built dataset provided by R. It contains measurements of diameter (inches), height (feet) and volume of timber (cubic feet) of 31 felled Black Cherry trees.

Note: the 'girth' variable within the dataset is actually the diameter of the tree. We can rename it now, which also adds it into our R Environment:

```
CODE
trees <- trees %>%
    rename(Diameter = Girth)

head(trees)
```

```
OUTPUT
  Diameter Height Volume
1      8.3     70   10.3
2      8.6     65   10.3
3      8.8     63   10.2
4     10.5     72   16.4
5     10.7     81   18.8
6     10.8     83   19.7
```

Original data source:

Meyer H. A. (1953). Forest Mensuration. Penns Valley Publishers.

Ryan T. A., Joiner B. L., Ryan B. F. (1976). The Minitab Student Handbook. Duxbury Press, North Scituate, MA. ISBN 087150359X

Figure 1: Black Cherry tree foliage and flowers, from Wikimedia Commons, by Sten Porse.

Analysis

Explore the data - EDA

For each variable we need to check the distribution, identify outliers and check for correlations between predictors (for MLR).

CODE

```
library(moments)

summary(trees)
```

OUTPUT

```
    Diameter        Height       Volume
 Min.   : 8.30   Min.   :63   Min.   :10.20
 1st Qu.:11.05   1st Qu.:72   1st Qu.:19.40
 Median :12.90   Median :76   Median :24.20
 Mean   :13.25   Mean   :76   Mean   :30.17
 3rd Qu.:15.25   3rd Qu.:80   3rd Qu.:37.30
 Max.   :20.60   Max.   :87   Max.   :77.00
```

CODE

```
# Apply the sd function for each column
apply(trees, 2, sd) # (margin = 2 means apply the function to columns)
```

OUTPUT

```
 Diameter    Height    Volume
 3.138139  6.371813 16.437846
```

CODE

```
skewness(trees)
```

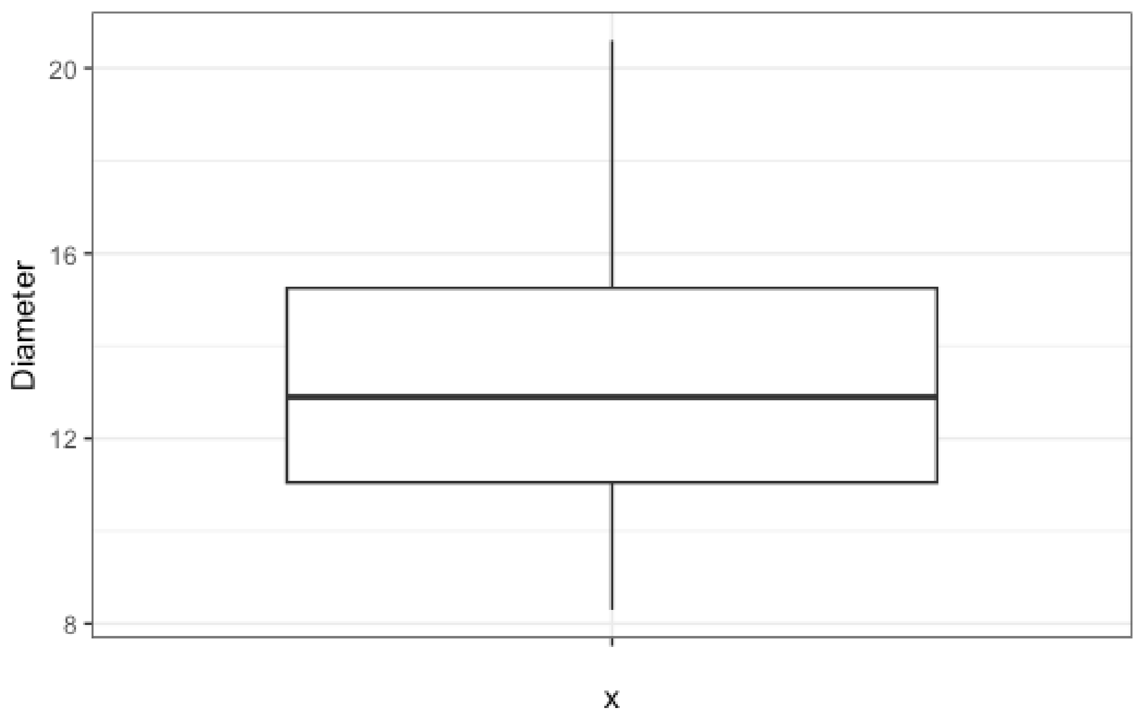
OUTPUT

```
  Diameter     Height     Volume
 0.5263163 -0.3748690  1.0643575
```

Diameter: The mean diameter is 13.25 inches and the median is 12.90 inches. The values range from 8.30 to 20.60 inches and the standard deviation is 3.14 inches. There is a slight skew, indicated by the skewness coefficient of 0.526. The slight skew is also visible in the following boxplot where the median lies closer to the first quartile.

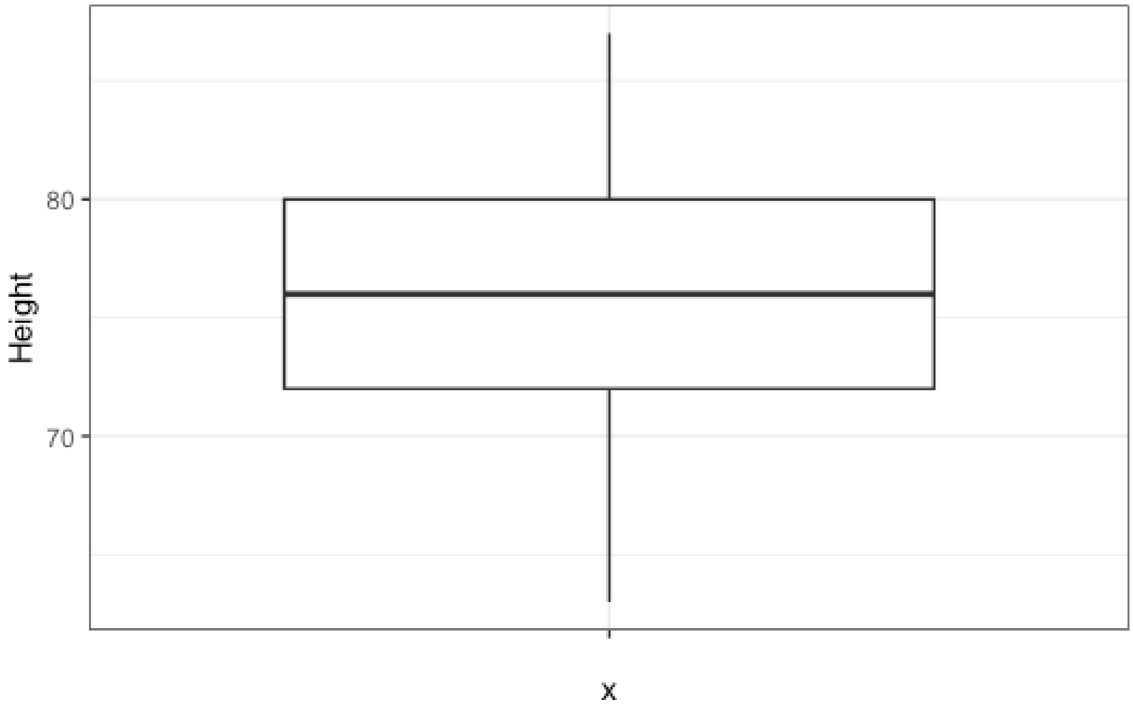
CODE

```
ggplot(trees, aes(x = "", y = Diameter)) +
    geom_boxplot() +
    theme_bw()
```



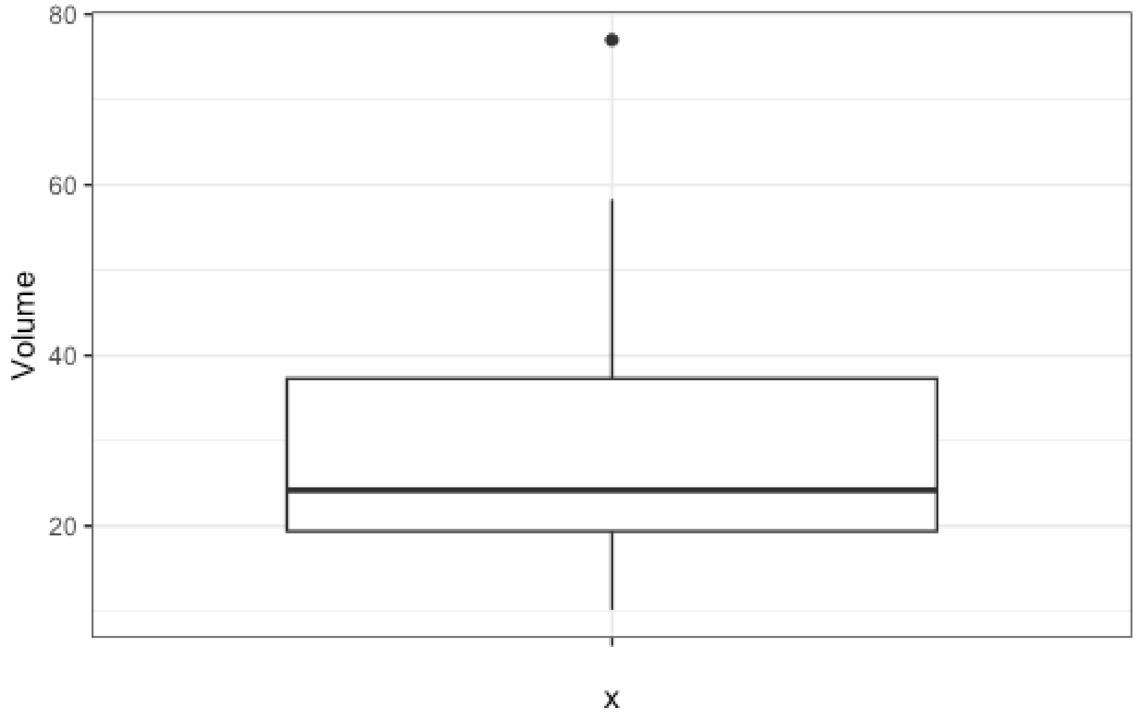
Height: The mean and median height is 76 feet and the values range from 63 to 87 feet. The standard deviation is 6.37 feet and the skewness coefficient is −0.37, indicating a relatively symmetrical distribution. The boxplot also shows a relatively symmetrical distribution, with the median lying close to the centre of the box.

```
CODE
ggplot(trees, aes(x = "", y = Height)) +
    geom_boxplot() +
    theme_bw()
```



Volume: The mean volume of timber is 30.17 and the median is 24.20 cubic feet. The values range from 10.20 to 77.00 cubic feet and the standard deviation is 16.44 cubic feet. The skewness coefficient is 1.06, indicating a the variable is positively skewed. This is also visible in the boxplot, where the median lies closer to the first quartile and there are some high values that are potential outliers.

```
CODE
ggplot(trees, aes(x = "", y = Volume)) +
    geom_boxplot() +
    theme_bw()
```



Why do the high values matter?

The high values would have high leverage in our model, causing the residuals to be skewed, violating our assumptions. The boxplot is therefore telling us that we might need to transform Volume to stabilise the variance of the regression, but it is worth confirming this with the diagnostic plots after fitting the model.

Explore the data - correlations

Correlations

CODE

```
cor(trees)
```

OUTPUT

```
          Diameter    Height    Volume
Diameter 1.0000000 0.5192801 0.9671194
Height   0.5192801 1.0000000 0.5982497
Volume   0.9671194 0.5982497 1.0000000
```

Why do we care about correlations between predictors?

multi-collinearity which may influence the model. If the relationship between these two variables was stronger, we would remove one of the variables to prevent this collinearity from affecting the model.

Note: For more information on collinearity and how it may impact the model, see Quinn & Keough p 127.

Implement the fitting process for each type of model

CODE

```
mod_slr <- lm(log10(Volume) ~ Diameter, data = trees)

mod_mlr <- lm(log10(Volume) ~ Diameter + Height, data = trees)
```

i Checkpoint

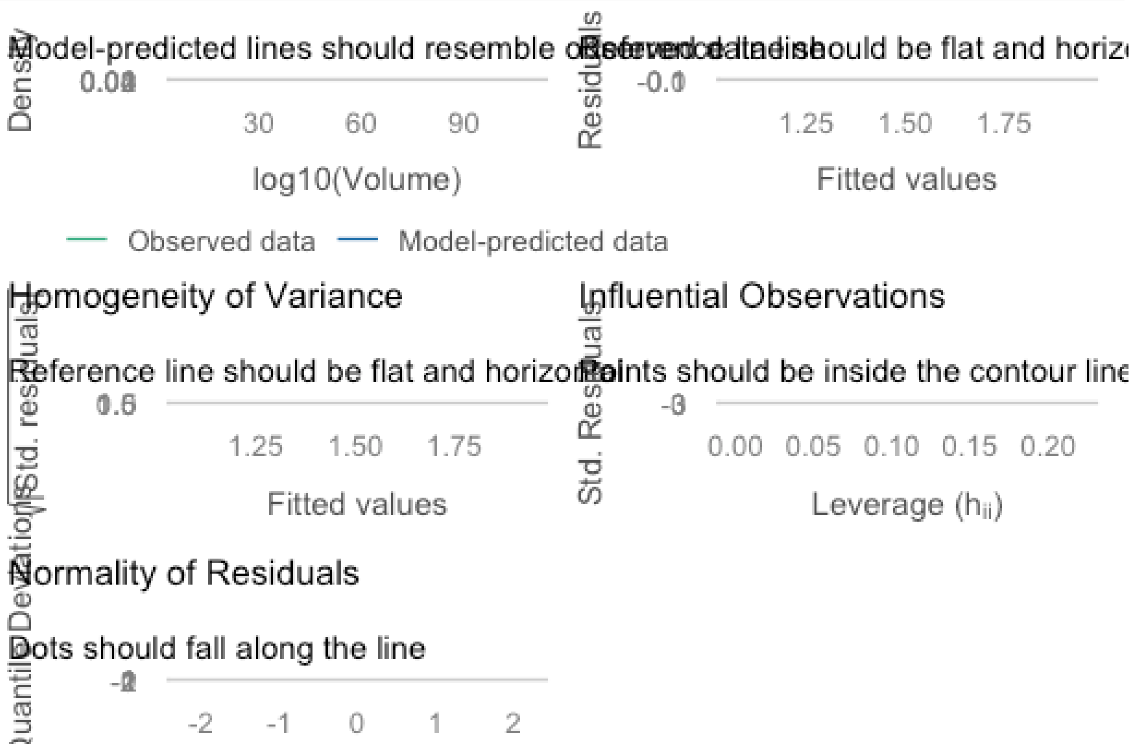
You should now be able to ...[PUT THIS AFTER EACH OBJECTIVE]

Test assumptions and transform if needed

CODE

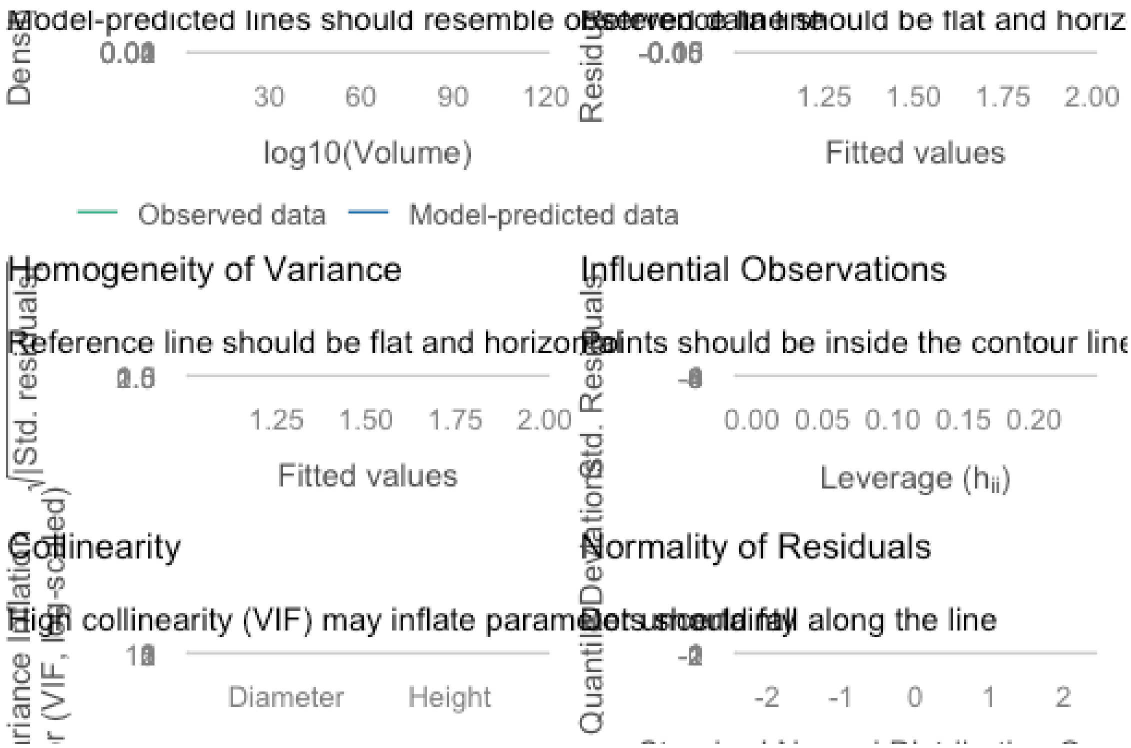
```
library(performance)

check_model(mod_slr)
```



CODE

```r
check_model(mod_mlr)
```



You do this to stabilise the variance of the regression to manage the leverage of the outliers in the variable. This reduces the skew. L10AREA is more likely to be a significant predictor.

> **i Checkpoint**
>
> You should now be able to …[PUT THIS AFTER EACH OBJECTIVE]

Evaluate model fit of SLR and MLR models.

CODE

```r
summary(mod_slr)
```

OUTPUT

```

Call:
lm(formula = log10(Volume) ~ Diameter, data = trees)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.09867 -0.04981  0.01255  0.03444  0.13218 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept) 0.485974   0.045176   10.76 1.23e-11 ***
Diameter    0.070601   0.003321   21.26  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.05708 on 29 degrees of freedom
Multiple R-squared:  0.9397,    Adjusted R-squared:  0.9376 
F-statistic:   452 on 1 and 29 DF,  p-value: < 2.2e-16
```

CODE

```
summary(mod_mlr)
```

OUTPUT

```

Call:
lm(formula = log10(Volume) ~ Diameter + Height, data = trees)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.076991 -0.037357 -0.004312  0.025586  0.073835 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.044552   0.093510   0.476    0.637    
Diameter    0.063098   0.002861  22.057  < 2e-16 ***
Height      0.007116   0.001409   5.051 2.41e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04202 on 28 degrees of freedom
Multiple R-squared:  0.9684,    Adjusted R-squared:  0.9662 
F-statistic: 429.7 on 2 and 28 DF,  p-value: < 2.2e-16
```

i Checkpoint

You should now be able to …[PUT THIS AFTER EACH OBJECTIVE]

Write up conclusions

Wrap-up

In this tutorial, we have walked through the processes for simple and multiple linear regression, including how to fit models, check assumptions and how to interpret the output. You will hopefully now be able to recall key differences between the two types of models, which will help direct your analysis.

You will now apply these skills in the lab. If you have any questions please reach out to your tutors.

Example exam questions

1. Describe 3 key differences between simple linear regression and multiple linear regression.
2. Describe how we would check the assumptions of a multiple linear regression. What would we do if we found that one of the assumptions was violated?
3. If you fitted a multiple linear regression and only one predictor was significant, would you still expect the model to be significant? Justify your answer (it may help to recall the hypotheses for MLR).